Weather App — Project Report

Course project, Project option 2: Weather app

1. What I built

A weather app in plain HTML, CSS and JavaScript. You can search for a city or use your GPS location, and it shows the current weather, the next 24 hours, and a 7-day forecast. I used two different weather websites (Open-Meteo and MET Norway) at the same time so the app pulls from two data sources, and I drew a small chart comparing their temperatures for the next 24 hours. There's also a favorites list and a °C/°F/K switch.

2. Tools used

HTML5 and CSS3 (flexbox + one media query for mobile), no CSS framework. Plain JavaScript, one file (`script.js`) — no React or any framework, as required. The `<canvas>` element for the comparison chart (just basic line drawing, no chart library). APIs: Open-Meteo for the main forecast + city search, and MET Norway / Yr.no as the second data source for comparison — both free, no API key needed, which made testing a lot easier. Python's `http.server` to run the page locally while testing (opening the HTML file directly caused some fetch requests to be blocked).

3. Directory structure

```
weather-app/
  index.html
  css/style.css
  js/script.js
  docs/report.md, report.html, report.pdf
```

I kept HTML, CSS and JS in separate files/folders so I wouldn't lose points there.

4. How many points I think this deserves

Feature	Max	I'd say	Why
PDF report	3	2–3	This document, kept it fairly short
Responsive (desktop + mobile)	4	3	One media query at 600px, only really tested on my own phone width in devtools
Works on Firefox/Safari/Edge/Chrome	3	2	Only tested in Chrome myself; used normal fetch/DOM stuff so it should work elsewhere too
Clear directory structure	2	2	css/js separated, see §3
Location search	1	1	Uses Open-Meteo's search API
GPS location	2	2	`navigator.geolocation`, just labeled "My Location" (no reverse geocoding)
2+ data sources	3	3	Open-Meteo + MET Norway, both actually called every time
3+ data sources	2	0	Didn't add a third one, wanted to keep it simple
Current weather	1	1	
24h hourly forecast	3	2–3	Hourly, but icon set only covers common weather codes
7-day forecast	3	3	
Icons everywhere	3	2	Icons show up but the mapping is pretty rough (a handful of buckets)
Look reflects the weather	2	2	Background color changes with temperature/rain
Two forecasts on one chart	3	2–3	Basic canvas line chart, no axis labels, shows both lines though
Favorites	2	2	Saved in localStorage, click to reload
°C/°F/K switch	2	2	
Total (being honest about the gaps)		**~30–33**	

I'm not 100% sure how strict the grading is on things like "does every icon look right" or "did you really test on Safari", so I left some of these as a range instead of a fixed number.

5. Known issues / things I didn't get to

- Only 2 weather sources, not 3 — didn't want to deal with API keys for a third one.
- The chart has no axis numbers, just two lines, so you can compare them but can't read exact values off it.
- Weather code → icon mapping isn't complete, some rarer codes just show a generic 🌡️ icon.
- GPS location just shows "My Location" instead of a real city name (would need another API call to reverse-geocode, skipped that).
- Favorites only remember the city name, not the coordinates, so clicking a favorite re-searches by name — if two cities share a name in different countries this could pick the wrong one.
- Not much error handling — if the internet drops mid-request the app might just get stuck on "Loading weather..." instead of showing a clean error.

6. How to run it

1. Open a terminal inside `weather-app/` and run `python -m http.server 8000` (or any other simple local server).
2. Open `http://localhost:8000` in your browser.
3. Search a city, or click "Use My Location".

7. AI usage declaration (mandatory)

I used **Claude (Anthropic)** to write this project. I described the assignment (course project brief, Project 2: Weather app) and asked for a working implementation, then asked for it to be rewritten in a simpler style after the first version came back far more polished/advanced than something I would realistically produce at my current skill level. The code, this report, and the testing were all generated by the AI; I did not hand-write any of it myself. No other AI tool was used.

Note: this project was built as a demonstration/test of what an AI assistant can produce, not submitted as anyone's real coursework.